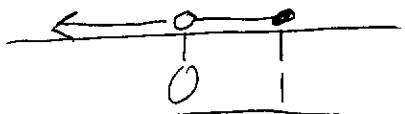
Review Test 2 1-10-18

(1a) $(f\cdot g)(x) = f(x)\cdot g(x) = \frac{1}{x^2}\cdot\sqrt{1-x} = \boxed{\frac{\sqrt{1-x}}{x^2}}$

$D = D_f \cap D_g = x \neq 0 \cap x \leq 1$

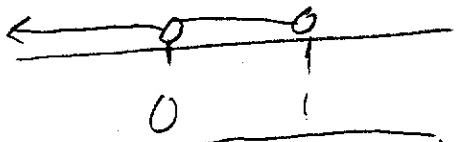


$\boxed{(-\infty,0)\cup(0,1]}$

$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} = \frac{\frac{1}{x^2}}{\sqrt{1-x}} = \boxed{\frac{1}{x^2\sqrt{1-x}}}$

$D = D_f \cap D_g$ also $D_g \neq 0$

$\sqrt{1-x} \neq 0$ so $x \neq 1$



$\boxed{(-\infty,0)\cup(0,1)}$

(1b) $\left(\frac{f}{g}\right)(-3) = \frac{1}{(-3)^2\sqrt{1-(-3)}} = \frac{1}{9\cdot 2} = \frac{1}{18}$

(1c) $(f\circ g)(x) = f(g(x)) = f(\sqrt{1-x}) = \frac{1}{(\sqrt{1-x})^2} = \boxed{\frac{1}{1-x}}$

$D: 1-x \geq 0 \cap 1-x \neq 0$

$1 \geq x \cap 1 \neq x$

$\boxed{(-\infty,1)}$

(1c) $(g\circ f)(x) = g(f(x)) = g\left(\frac{1}{x^2}\right) = \sqrt{1-\frac{1}{x^2}} = \sqrt{\frac{x^2}{x^2}-\frac{1}{x^2}} =$

$\sqrt{\frac{x^2-1}{x^2}} = \frac{\sqrt{x^2-1}}{\sqrt{x^2}} = \boxed{\frac{\sqrt{x^2-1}}{|x|}}$

D: $x \neq 0 \wedge x^2-1 \geq 0$

$(x+1)(x-1) \geq 0$

$\oplus \quad \ominus \quad \oplus$

$-1 \qquad 1$

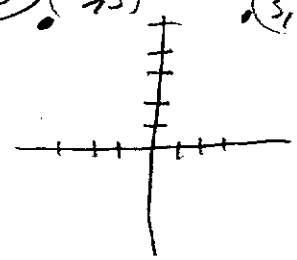
$\boxed{(-\infty,-1] \cap [1,\infty)}$

(1d) $(f\circ g)(0) = \frac{1}{1-0} = \boxed{1}$

(2) $g(x) = x^2+4$ Many correct answers are possible.

$f(x) = x^{3/2}$

(3) $(-3,5)$ $(3,5)$



(3a) $(g\circ f)(-3) = g(f(-3)) = g(5)$

$g(5) = \sqrt{5^2-16} = \sqrt{9} = \boxed{3}$

(3b) $(f\circ g)(5) = f(g(5)) = f(\sqrt{5^2-16})$

$= f(3) = \boxed{5}$ from graph

(4) $f(f(x)) = f\left(\frac{2}{x-1}\right) = \frac{2}{\frac{2}{x-1}-1} =$

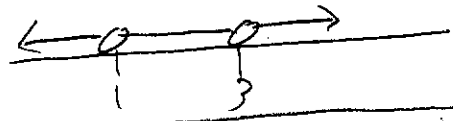
$= \left(\frac{2}{\frac{2}{x-1}-1}\right)\left(\frac{x-1}{x-1}\right) = \frac{2(x-1)}{2-1(x-1)} = \frac{2x-2}{2-x+1} =$

$= \frac{2x-2}{-x+3} = \boxed{\frac{2x-2}{3-x}}$

Domain is determined by

$f\left(\frac{2}{x-1}\right)$ and $\frac{2x-2}{3-x}$

$x \neq 1$ and $x \neq 3$



$\boxed{(-\infty,1) \cup (1,3) \cup (3,\infty)}$

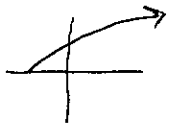
(5a) One-to-one: You can solve for x to see if there is only one y or you can graph and use the horizontal line test.

ie. $f(x) = x^2$

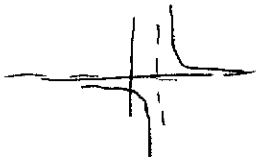
$y = x^2$

$x = \pm\sqrt{y}$ since one x yields two different ys, not 1-1.

or graph fails hor. line test.

(5a) $f(x) = \sqrt{x+5}$  so $\boxed{1\text{-}1}$

(5b) The graph is a parabola so $\boxed{\text{not } 1\text{-}1}$.

(5c) $h(x) = \dfrac{1}{x-1}$  so $\boxed{1\text{-}1}$

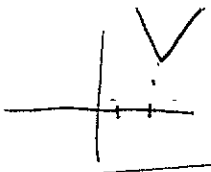
or $y = \dfrac{1}{x-1}$ solve for x

$y(x-1) = 1$

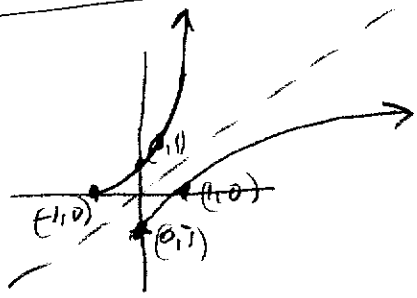
$xy - y = 1$

$xy = 1 + y$

$x = \dfrac{1+y}{y}$ so for each x, only one y, so $\boxed{1\text{-}1}$

(5d)  $\boxed{\text{not } 1\text{-}1}$

(6)



(7) $f: (4,5)$ so $f^{-1}: (5,4)$

$(g \circ f^{-1})(5) = g(f^{-1}(5)) = g(4) = 2(4) - 6 = \boxed{2}$

(8) $(f \circ g)(x) = (g \circ f)(x)$

$$f(g(x)) = f(x^3-3) = \sqrt[3]{(x^3-3)+3} = \sqrt[3]{x^3} = x$$

$$g(f(x)) = g(\sqrt[3]{x+3}) = (\sqrt[3]{x+3})^3 - 3 = x+3-3 = x$$

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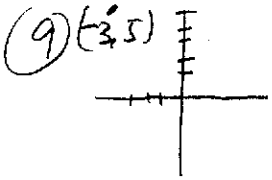
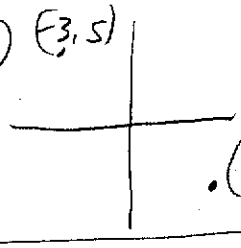
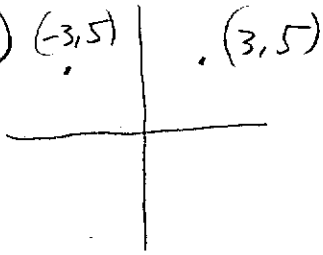
(9) 

(9a) 

so $f(3) = \boxed{-5}$

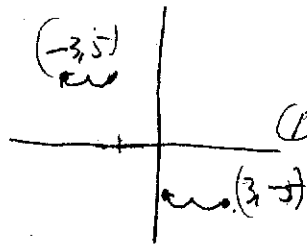
since f has $(3,-5)$, f^{-1} has $(-5,3)$

so $f^{-1}(-5) = \boxed{3}$

(9b) 

$f(3) = \boxed{5}$

Since f is not 1-1, it does not have an inverse function, so $f^{-1}(5)$ does not exist. There is an inverse relation, but not an inverse function.

(9c) 

$y = -2f(x+2) - 1$

(1) $y = f(x+2)$ = move points left 2, to get $(-5,5)$ and $(1,-5)$.

(2) $y = -2f(x+2)$, now multiply the y-coordinates by -2 to get $(-5,-10)$ and $(1,10)$.

(3) $y = -2f(x+2) - 1$, now move y-coordinates down 1. to get $\boxed{(-5,-11)}$ and $\boxed{(1,9)}$.

(9d) Since even you have $(-3,5)$ and $(3,5)$

$y=f(-x)$ is a reflection across y-axis: $(3,5)$ and $(-3,5)$

$y=f(-x+2)$ is $\rightarrow 2$ thus : $(5,5)$ and $(-1,5)$

$y=2f(-x+2)$ multiplies they by 2 : $(5,10)$ and $(-1,10)$

$y=2f(-x+2)-1$ is $\downarrow 1$: $\boxed{(5,9)}$ and $\boxed{(-1,9)}$

(10a) $y=\dfrac{x-5}{x+2}$

$x=\dfrac{y-5}{y+2}$

$x(y+2)=y-5$

$xy+2x=y-5$

$xy-y=-2x-5$

$y(x-1)=-2x-5$

$y=\dfrac{-2x-5}{x-1}$

$\boxed{f^{-1}(x)=\dfrac{-2x-5}{x-1}}$

$D_{f^{-1}}$: $x\neq 1$

$\boxed{(-\infty,1)\cup(1,\infty)}$

(10b) $y=\sqrt{x+1}+2$ Note: D_f is

$x=\sqrt{y+1}+2$

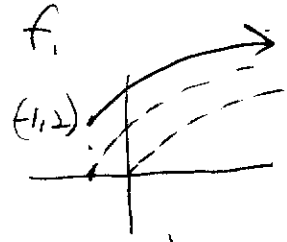
$\sqrt{y+1}=2-x$

$y+1=(2-x)^2$

$y+1=4-4x+x^2$

$y=x^2-4x+3$ This in not 1-1, so find range of f.

$\boxed{f^{-1}(x)=x^2-4x+3,\ x\geq 2}$



R_f: $[2,\infty)$ so

$D_{f^{-1}}$ is $[2,\infty)$

(10c) $y = x^2 - 2x, \quad x \geq 1$

$x = y^2 - 2y, \quad y \geq 1$

$y^2 - 2y = x \qquad y \geq 1$

Complete the square on left side.

$y^2 - 2y + 1 = x + 1$

$\boxed{\frac{1}{2}(-2)}^2$

$(y-1)^2 = x+1 \qquad y \geq 1$

$y - 1 = \pm\sqrt{x+1}$ since the left side $y-1$ is $\oplus$ so from the $\pm$ only the $+$.

$y - 1 = \sqrt{x+1}$

$\boxed{y = \sqrt{x+1} + 1}$

D: $x + 1 \geq 0$

$x \geq -1$

D: $\boxed{[-1, \infty)}$

(10d) $y = x^2 - 2x, \; x \le 1$

$x = y^2 - 2y, \quad y \le 1$

$y^2 - 2y = x \quad y \le 1$

complete the square on the left side.

$y^2 - 2y + 1 = x + 1$

$\boxed{\tfrac{1}{2}(-2)}^2 = (-1)^2 = 1$

$(y-1)^2 = x+1, \quad y \le 1$

$y - 1 = \pm\sqrt{x+1}$ since $y \le 1$ makes $y-1$ negative so only $(-)$

$y - 1 = -\sqrt{x+1}$

$\boxed{y = -\sqrt{x+1} + 1}$ D: $x + 1 \ge 0$

$x \ge -1$ thus $\boxed{[-1, \infty)}$ D

or $\boxed{y = 1 - \sqrt{x+1}}$

(10e) $y = \frac{1}{x}$

$x = \frac{1}{y}$

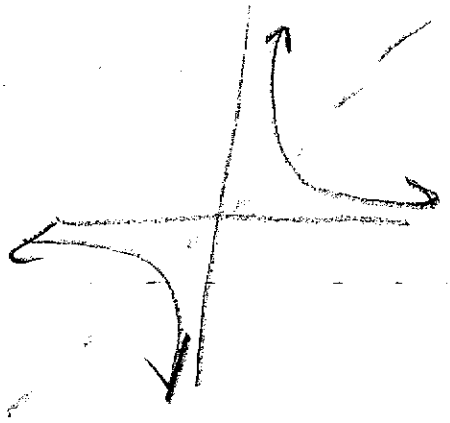
$y(x) = \left(\frac{1}{y}\right)y$

$\frac{1}{x}(xy) = (1)\frac{1}{x}$

$y = \frac{1}{x}$

so $f^{-1}(x) = \frac{1}{x}$ $\quad D: (-\infty, 0) \cup (0, \infty)$

Seems strange that $\frac{1}{x}$ is the inverse of itself, but look at the graph of $\frac{1}{x}$. It is mirrored across the line $y = x$.



(11) $f(x) = \dfrac{x-5}{x+2}$, since $f^{-1}(x) = \dfrac{-2x-5}{x-1}$ has $D_{f^{-1}}: (-\infty,1)\cup(1,\infty)$

The range of f is $\boxed{(-\infty,1)\cup(1,\infty)}$.

The HA of f is 1, so no it doesn't cross its HA. You can tell by looking at the range of f: $(-\infty,1)\cup(1,\infty)$.

(12a) $2+\sqrt{5}i$ $\boxed{2-\sqrt{5}i}$

(12b) $-3i = 0-3i$ conjugate: $0+3i$ $\boxed{3i}$

(12c) $25 = 25+0i$ conjugate $\boxed{25-0i}$

(13a) $\sqrt{(3+4i)(4i-3)} = \sqrt{12i-9+16i^2-12i} = \sqrt{-9+16(-1)} = \sqrt{-25}$
$= \boxed{5i}$

(13b) $\sqrt{-20}\sqrt{-80} = \sqrt{20}\,i \cdot \sqrt{80}\,i = 2\sqrt{5}\cdot 4\sqrt{5}\,i^2 = -2\cdot 4\cdot 5 = \boxed{-40}$

$\uparrow$ $16\cdot 5$

(13c) $\sqrt{-20} - \sqrt{-80} = \sqrt{20}\,i - \sqrt{80}\,i = 2\sqrt{5}\,i - 4\sqrt{5}\,i = \boxed{-2\sqrt{5}\,i}$

(14a) $(4-3i)-(4+3i) = 4-3i-4-3i = \boxed{-6i = 0-6i}$

(14b) $(i^{23}-1)^2 = (-i-1)^2 = (-i)^2 - 2(1)(-i) + 1^2 = i^2+2i+1 =$
$-1+2i+1 = \boxed{2i}$

or $\boxed{0+2i}$

note: $i^{23} = (i^4)^5 i^3 = 1^5 \cdot i^2 \cdot i = 1\cdot(-1)\cdot i = -i$

(14c) $\dfrac{1+i}{i^3} = \dfrac{1+i}{-i} = \left(\dfrac{1+i}{-i}\right)\dfrac{i}{i} = \dfrac{i+i^2}{-i^2} = \dfrac{i+(-1)}{-(-1)} = \dfrac{-1+i}{1} = \boxed{-1+i}$

note: $i^3 = -i$

(14d) $\left(\frac{1-i}{3+2i}\right)\left(\frac{3-2i}{3-2i}\right) = \frac{3-2i-3i+2i^2}{9+4} = \frac{3-5i-2}{13} = \frac{1-5i}{13}$

$= \boxed{\frac{1}{13} - \frac{5}{13}i}$

(15a) $4x(x+3) = -9$

$4x^2 + 12x + 9 = 0$

$b^2 - 4ac \rightarrow 144 - 4(4)9) = 0$ so one real solution.

$x = \frac{-12 \pm \sqrt{12^2 - 4(4)(9)}}{2(4)} = \frac{-12 \pm 0}{8} = \boxed{-\frac{3}{2}}$

(15b) $3x^2 - 4x + 5 = 0$

$b^2 - 4ac \rightarrow (-4)^2 - 4(3)(5) =$ negative, thus two complex solutions.

$x = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(3)5}}{2(3)} = \frac{4 \pm \sqrt{16-60}}{6} = \frac{4 \pm \sqrt{-44}}{6} = \frac{4 \pm 2\sqrt{11}}{6}$

$= \frac{2 \pm \sqrt{11}}{3} = \boxed{\frac{2}{3} \pm \frac{\sqrt{11}i}{3} = \frac{2 \pm \sqrt{11}i}{3}}$

(16) $f(x) = -2x^2 + 8x - 8$

$-\frac{b}{2a} \rightarrow \frac{-8}{2(-2)} = 2$

$f(2) = -2 \cdot 4 + 16 - 8 = 0$

thus $V(2,0)$

$-2x^2 + 8x - 8 = 0$

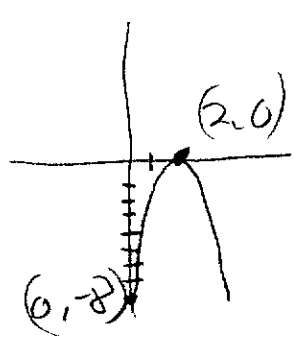
$x^2 - 4x + 4 = 0$

$(x-2)^2 = 0$

$x = 2$

thus $(2,0)$ is x-intercept

x	y
0	-8



(16b) $g(x) = x^2 - x - 6$

$\frac{-b}{2a} \rightarrow \frac{-(-1)}{2} = \frac{1}{2}$

$f(\frac{1}{2}) = \frac{1}{4} - \frac{1}{2} - 6 =$

$\frac{1}{4} - \frac{2}{4} - 6 = -6\frac{1}{4} = -\frac{25}{4}$

thus $V(\frac{1}{2}, \frac{-25}{4})$

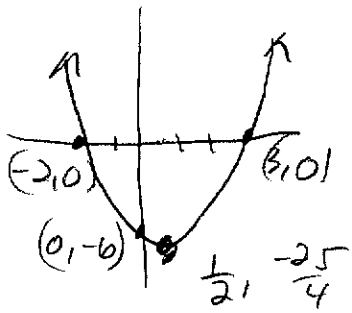
x-intercepts

$x^2 - x - 6 = 0$

$(x-3)(x+2) = 0$

$x = 3 \quad x = -2$

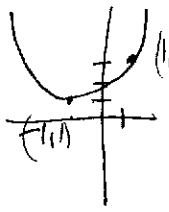
thus $(3,0)$ $(-2,0)$



y-int

$y = -6$

(17a)



No x-intercepts.
Range $[1, \infty)$

(17b) $y = a(x-h)^2 + k$

Plug in $(1,3)$ into (x,y) and $(-1,1)$ into (h,k), solve for a.

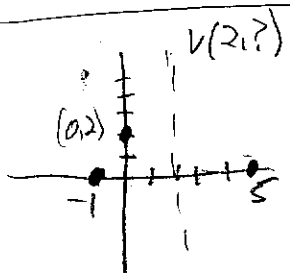
$3 = a(1-(-1))^2 + 1$

$3 = 4a + 1$

$4a = 2$

$a = \frac{1}{2}$ thus $f(x) = \frac{1}{2}(x+1)^2 + 1$

(18)



Find a

$y = a(x-(-1))(x-5)$

$y = a(x+1)(x-5)$

sub in $(x,y) \rightarrow (0,2)$

$2 = a(0+1)(0-5)$

$2 = a(1)(-5)$

$a = -\frac{2}{5}$ now find k.

$y = a(x-h)^2 + k$ plug in $V(2,k)$ $(0,2)$

$2 = -\frac{2}{5}(0-2)^2 + k$

$2 = -\frac{2}{5}(4) + k$

$k = 2 + \frac{8}{5} = \frac{18}{5}$

$R: (-\infty, \frac{18}{5}]$

(19a) Find y-coor of vertex.

$-\frac{b}{2a} \rightarrow \frac{-80}{2(-16)} = \frac{40}{16} = \frac{5}{2} = 2\frac{1}{2}$ sec

$h\left(\frac{5}{2}\right) = -16\left(\frac{5}{2}\right)^2 + 80\left(\frac{5}{2}\right) + 96$

$= -100 + 296 = \boxed{196 \text{ ft}}$ max height

(19b) Let $h = 0$

$-16t^2 + 80t + 96 = 0 \quad \left(\frac{-1}{16}\right)$

$t^2 - 5t - 6 = 0$

$(t-6)(t+1) = 0$

$\boxed{t = 6 \text{ sec}}$ $t = -1$ sec

(19c) $-16t^2 + 80t + 96 > 160$

$-16t^2 + 80t - 64 > 0$

$-\frac{1}{16}\left[-16t^2 + 80t - 64 > 0\right]$

$t^2 - 5t + 4 < 0$

$(t-4)(t-1) < 0$

$- - \mid - + \mid + +$

$\oplus \quad \ominus \quad \oplus$

$1 \qquad 4$

thus $\boxed{1 < t < 4 \text{ sec}}$

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(20) Revenue = price × number sold

$R = (40 - 0.2x)x = -.02x^2 + 40x$

$R(x) = -0.2x^2 + 40x$ Find vertex. $\frac{-b}{2a} \rightarrow \frac{-40}{2(-0.2)} = \frac{20}{\left(\frac{2}{10}\right)} = \frac{200}{2} = 100 =$ x coor of vertex

Now use $x = 100$ in $p = 40 - 0.2x$

$p = 40 - 0.2(100)$

$p = 40 - 20 = \boxed{\$20 \text{ price}}$

Max revenue: Find y-coor of vertex

$R(100) = -0.2(100)^2 + 40(100) = -0.2(10{,}000) + 4000$

$= -2000 + 4000 = \boxed{\$2000 \text{ max revenue}}$

Let x = # sold

(21a) Profit = Revenue − cost

Revenue = price × number sold $= \left(-\frac{25}{4}x + 1100\right)x = -\frac{25}{4}x^2 + 1100x$

Profit $= -\frac{25}{4}x^2 + 1100x - (7500 + 500x) = \boxed{-\frac{25}{4}x^2 + 600x - 7500} = P(x)$

(21b) Find x-coor of vertex

$\frac{-600}{2\left(-\frac{25}{4}\right)} = -600\left(\frac{4}{2(-25)}\right) = \boxed{48 \text{ units}}$

$$\begin{array}{r} 25 \\ \underline{12} \\ 50 \\ \underline{5} \\ 250 \\ \underline{25} \\ 300 \end{array}$$

(21c) Use $p = -\frac{25}{4}x + 1100$

$p(48) = -\frac{25}{4}(48) + 1100$

$= -25(12) + 1100 = \boxed{\$800}$

(22) Directions should have said find all real zeros.

$f(x) = 2(x^3 - 9x)(x+3)^3(x^2+4)^3 = x(x+3)(x-3)(x+3)^3(x^2+4)^3$

zeros: $x = 0, -3, 3$

multiplicities: one, four, one

$x^3 - 9x = 0$

$x(x^2 - 9) = 0$

$x(x+3)(x-3) = 0$

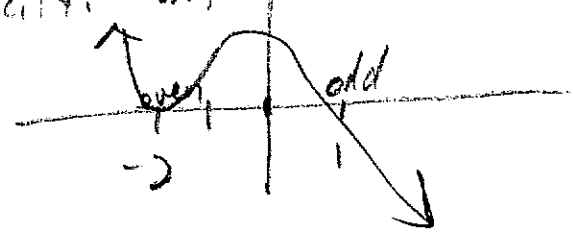
zeros: $x = 0, -3, 3$

(23) $f(x) = -2(x-1)(x+2)^2$

leading term: $-2x^3$ ↑ ↓

zeros: $x = 1, -2$

multiplicity: odd, even



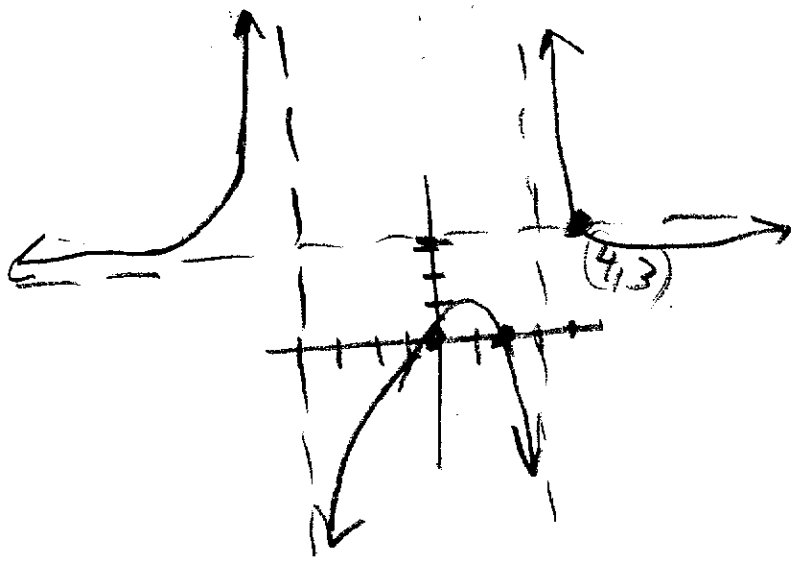
optional: − − + | − − + | − + +

(−) (+) (−)

-2 1

(24a) $f(x) = \dfrac{3x^2-6x}{x^2+x-12}$

$f(x) = \dfrac{3x(x-2)}{(x+4)(x-3)}$ D: $x \neq -4, 3$



VA: $x=-4$, $x=3$

No holes.

HA. $\dfrac{3x^2}{x^2} = 3$ so $y=3$

Does it cross the HA?

$$\frac{3x^2-6x}{x^2+x-12} = 3$$

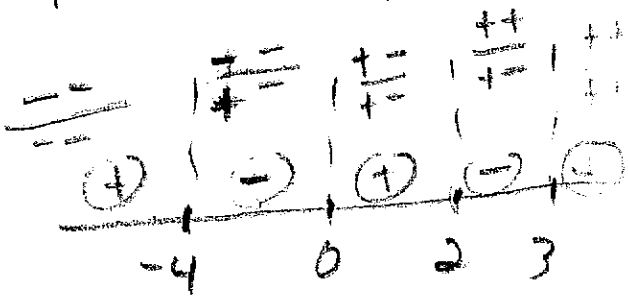
$$3x^2-6x = 3x^2+3x-36$$

$$36 = 9x$$

$$x = 4$$

yes, at (4,3)

symmetry? Look at graph so far and you can see there is no sym.



$$\frac{3x(x-2)}{(x+4)(x-3)}$$

x and y-intercepts

Let $y=0$ $\dfrac{3x(x-2)}{(x+4)(x-3)} = 0$

x	y
2	0
0	0

$x=2$, $x=0$

Let $x=0$

$$f(0) = \frac{3(0^2)-6(0)}{-12} = \frac{0}{-12} = 0$$

(246) $f(x) = \dfrac{(x+3)(x+1)}{(x-2)(x+1)(x-2)}$, D: $x = 2, -1$

$$f(x) = \frac{x+3}{(x-2)^2}$$

VA: $x = 2$

Holes: $x = -1$

$f(-1) = \dfrac{-1+3}{(-1-2)^2} = \dfrac{2}{9}$ thus $(-1, \frac{2}{9})$ hole

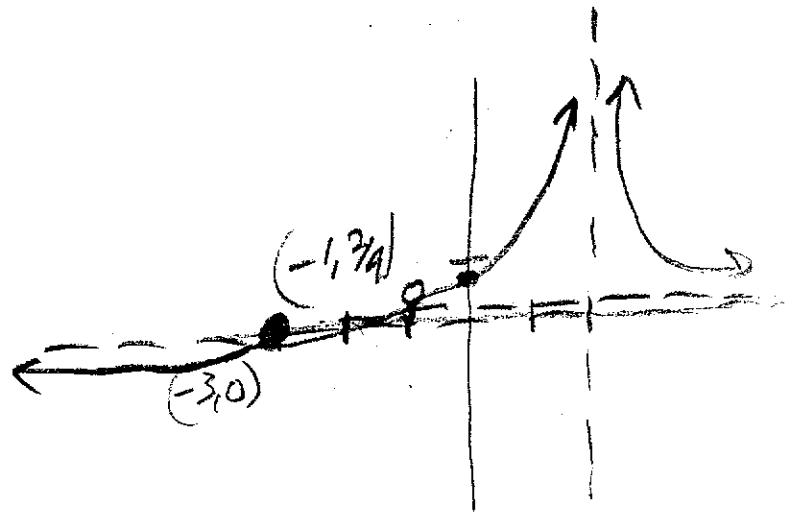
HA: $\dfrac{x^2}{x^3} = \dfrac{1}{x}$ $y = 0$ is HA

Cross HA?

$\dfrac{x+3}{(x-2)^2} = 0$ yes, at $(-3, 0)$

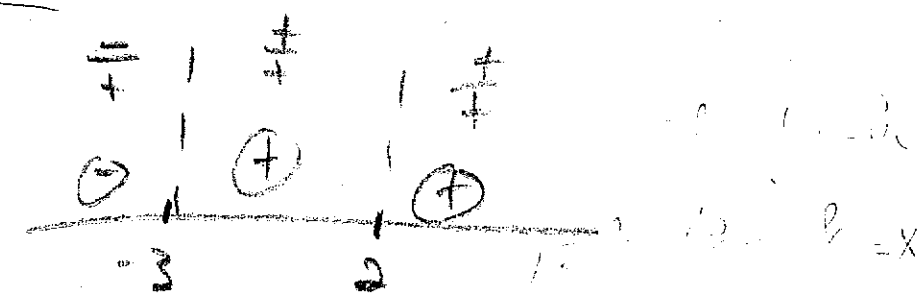
x	y
-3	0
0	$3/4$

$\dfrac{0+3}{(0-2)^2} = \dfrac{3}{4}$



Sym - none

$f(x) = \dfrac{x+3}{(x-2)^2} \longrightarrow$



(25)

$$\begin{array}{r|rrrr} 3 & 2 & 0 & 4 & -5 \\ & & 6 & 18 & 66 \\ \hline & 2 & 6 & 22 & 61 \\ & x^2 & x & c & r \end{array} \rightarrow 2x^2+6x+22+\frac{61}{x-3}$$

so $\dfrac{2x^3+4x-5}{x-3} = \boxed{2x^2+6x+22+\dfrac{61}{x-3}}$

(26)

$$\begin{array}{r|rrrrrr} 5 & 3 & -14 & 0 & -22 & -7 & -13 \\ & & 15 & 5 & 25 & 15 & 40 \\ \hline & 3 & 1 & 5 & 3 & 8 & 27 \end{array}$$

No since $r = 27$ not 0.

(27) $f(4) = 4(4)^4 - 3(4)^3 + 2(4)^2 - 5(4) + 8$

$= 1024 - 192 + 32 - 20 + 8 = 1064 - 212 \neq 0$

So not a factor.

$4 \cdot 4 \cdot 4 \cdot 4 \cdot 4$

$64 \times 16 = 384 + 64 = 1024$

$64 \times 3 = 192$

(28a) $-3x^2+5x-2\geq 0$

$(-3x+2)(x-1)\geq 0$

$\frac{2}{3}$ $\quad$ 1

$\boxed{\left[\frac{2}{3}, 1\right]}$

28b) $6x^3(x-1)^2-12x^2(x-1)<0$

$6x^2(x-1)[x(x-1)-2]<0$

$6x^2(x-1)(x^2-x-2)<0$

$6x^2(x-1)(x-2)(x+1)<0$

-1 $\quad$ 0 $\quad$ 1 $\quad$ 2

$\boxed{(-\infty,-1)\cup(1,2)}$

28c) $\frac{x+4}{2-x}-3<0$

$\frac{x+4}{2-x}-\frac{3(2-x)}{(2-x)}<0$

$\frac{x+4-6+3x}{2-x}<0$

$\frac{4x-2}{2-x}<0$

$\frac{1}{2}$ $\quad$ 2 undefined

$\boxed{\left(-\infty,\frac{1}{2}\right)\cup(2,\infty)}$

28d) $\frac{x-3}{x+3}-\frac{x+3}{x-3}\leq 0$

$\left(\frac{x-3}{x+3}\right)\left(\frac{x-3}{x-3}\right)-\left(\frac{x+3}{x-3}\right)\left(\frac{x+3}{x+3}\right)\leq 0$

$\frac{x^2-6x+9-(x^2+6x+9)}{(x+3)(x-3)}\leq 0$

$\frac{-12x}{(x+3)(x-3)}\leq 0$

-3 undefined $\quad$ 0 $\quad$ 3 undefined

$\boxed{(-3,0]\cup(3,\infty)}$